Sessional Quiz **Date: 09/12/2024**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

L-2/T-II CSE 220 Signals and Linear Systems Sessional

Time: 60 mins **Marks: 40**

Student Name: ______________________________ Student ID: ____________

> There are 8 questions in total. Answer all the questions in the spaces provided on the question sheets. No extra sheets will be provided

1. **What is the output of the following Python program?** (5)

```
import numpy as np

data = np.array([3, 6, 1, 2, 4, 7, 8])
data = data[ data % 2 == 0 ]

data = data / 2
result = data ** 2

print(result.sum())
```

Write down the Output below:

2. **What is the output of the following Python program?** (5)

```python
projects = {
"Project_A": {"priority": 1, "team_performance": 85, "deadline": 15},
"Project_B": {"priority": 2, "team_performance": 70, "deadline": 20},
"Project_C": {"priority": 3, "team_performance": 90, "deadline": 10},
"Project_D": {"priority": 1, "team_performance": 60, "deadline": 25},
"Project_E": {"priority": 3, "team_performance": 75, "deadline": 18},
"Project_F": {"priority": 3, "team_performance": 80, "deadline": 12},
"Project_G": {"priority": 1, "team_performance": 88, "deadline": 8},
"Project_H": {"priority": 1, "team_performance": 65, "deadline": 30},
}

selected_projects = list(projects.items())[1::3]
total_resources = 0

for i, (project, details) in enumerate(selected_projects):
    allocated_resources = 0

    if details["priority"] == 1:
        allocated_resources += 50
    elif details["priority"] == 2:
        allocated_resources += 30
    else:
        allocated_resources += 20

    if details["team_performance"] > 80:
        allocated_resources += 20
    elif details["team_performance"] > 60:
        allocated_resources += 10

    if i % 2 == 0:
        allocated_resources += 10

    if details["deadline"] <= 15:
        allocated_resources += 15
    else:
        allocated_resources += 5

    total_resources += allocated_resources
    print(f"Resources allocated for {project}: {allocated_resources}")

total_resources += 50

print("Total allocated resources:", total_resources)
```

<u>**Write down the Output below:**</u>

3. A temperature sensor in a remote location collects data samples of the temperature at a rate of **3 samples per second**, forming a discrete-time signal $x[n]$, where n represents the sample index. The data needs to be transmitted to a central monitoring system via a low-bandwidth satellite link that can only handle a maximum rate of **2 samples per second**. As a result, the exact signal cannot be transmitted to the central server. But the central system requires the data at the original rate of **3 samples per second** for accurate analysis and visualization. (5)

   Question I: Explain a way to send the signal through the satellite link and reconstruct the signal at the central server. The reconstructed signal will be an approximation of the original signal.

   Question II: For the discrete-time signal over the first 3 seconds:

   $$x[n] = \{0.0, 0.5, 0.87, 1.0, 0.87, 0.5\}, \quad n = 0, 1, 2, 3, 4, 5$$

   1. What is the **input signal to the channel**?
   2. What is the **reconstructed signal** at the receiver ?

4. Suppose you have three signals `a[n]`, `b[n]` and `c[n]` defined below. We want to generate three signals `x[n]`, `y[n]` and `z[n]`. Now determine if it is possible to express each of `x[n]`, `y[n]` and `z[n]` as the result of convolution between any two signals among `a[n]`, `b[n]` and `c[n]` (e.g. `a * b`). If it is, then express them as a result of convolution. Otherwise, explicitly mention that it is not possible. Assume that the signal values are padded with zeroes. (5)

$$a[n] = [1, 0, 1, 1]$$
$$b[n] = [1, 1, 0, 1]$$
$$c[n] = [1, 1, 1, 1]$$

$$x[n] = [1, 1, 1, 3, 1, 1, 1]$$
$$y[n] = [1, 1, 2, 3, 2, 2, 1]$$
$$z[n] = [1, 2, 1, 2, 2, 0, 1]$$

5. Two windmills, **Windmill A** and **Windmill B**, generate power based on the wind's direction (x): (5)

- **Windmill A**: Generates power that depends only on the **magnitude of the wind speed** and is unaffected by changes in the wind's direction.
- **Windmill B**: Generates power that depends on both the **speed and direction of the wind** and flips sign when the wind direction reverses.

The total power generated by the two windmills is modeled as:

$$P(x) = x^3 + 3\cos(x) - 5x^2 + \sin(2x).$$

Question 1: Separate and Verify Power Contributions

1. **Separate the Contributions:** Decompose $P(x)$ into its **even** and **odd** components:
   - The **even component** corresponds to the contribution from **Windmill A** (unaffected by wind direction reversal).
   - The **odd component** corresponds to the contribution from **Windmill B** (flips sign with wind direction reversal).
2. **Verify:** Show that the combined power $P(x)$ can be reconstructed by summing the even and odd components.

Question 2: Fourier Series Analysis

1. **For Windmill A (Even Component):** Which Fourier series coefficients (A_n or B_n) will be zero for its power contribution? Explain.
2. **For Windmill B (Odd Component):** Which Fourier series coefficients (A_n or B_n) will be zero for its power contribution? Explain.

6. Find the fourier transform of e^{-t^2}. After that find the energy in time domain and frequency domain and show that they are equal. (5)

7. (a) The result of cross-correlation between 2 signals is given as `r[n] = [21.7, -29.2, 23.5, -27.6]`. You do not know the two original signals but you know that they are out of phase with respect to each other. Given all these information, find out the sample lag. (2)

(b) Suppose, two discrete signals `x[n] = [1, -2]` and `y[n] = [2.5, -2.7, 3.4]` are given. Calculate the frequency spectrum of the circular convolution of x[n] & y[n] using the 4-point Fast Fourier Transform (FFT). (3)

8. (a) You are given implementations of a CFT and an DFT algorithm. Is it possible to perform Laplace and z-transform using the given implementations without changing their code? If yes, describe how. If no, explain why. (5)

(b) You are given the following function defined on the set of integers:

$$x[n] = \begin{cases} 1, \text{ if } n \geq 0 \\ 0, \text{ otherwise} \end{cases}$$

Then formulate the function $x[-n-1]$ and draw the plot.

9. (Bonus) Please write down some of the biggest challenges you faced in this course. Kindly suggest some ways that you think will be helpful for improving this course. (3)